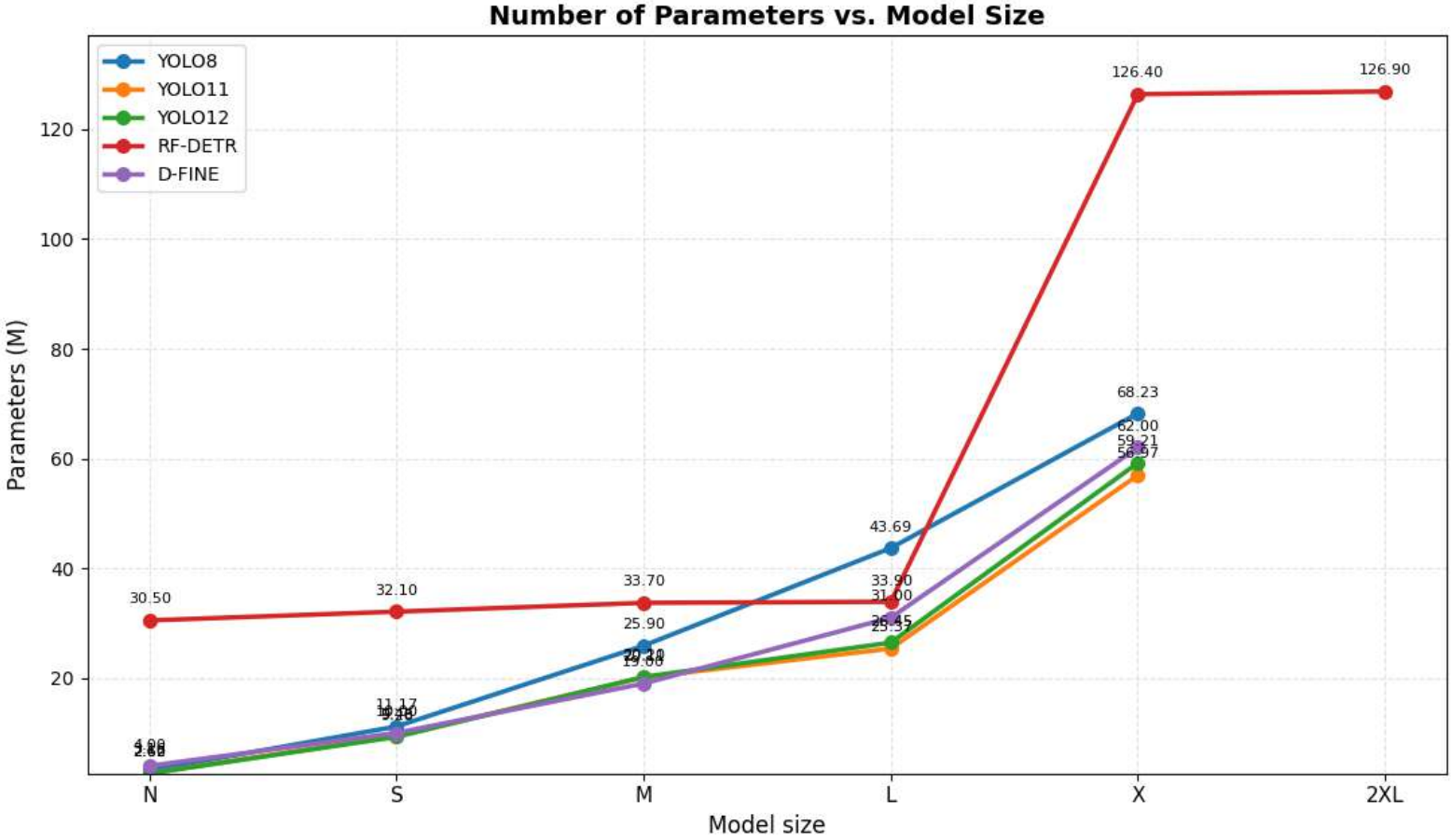
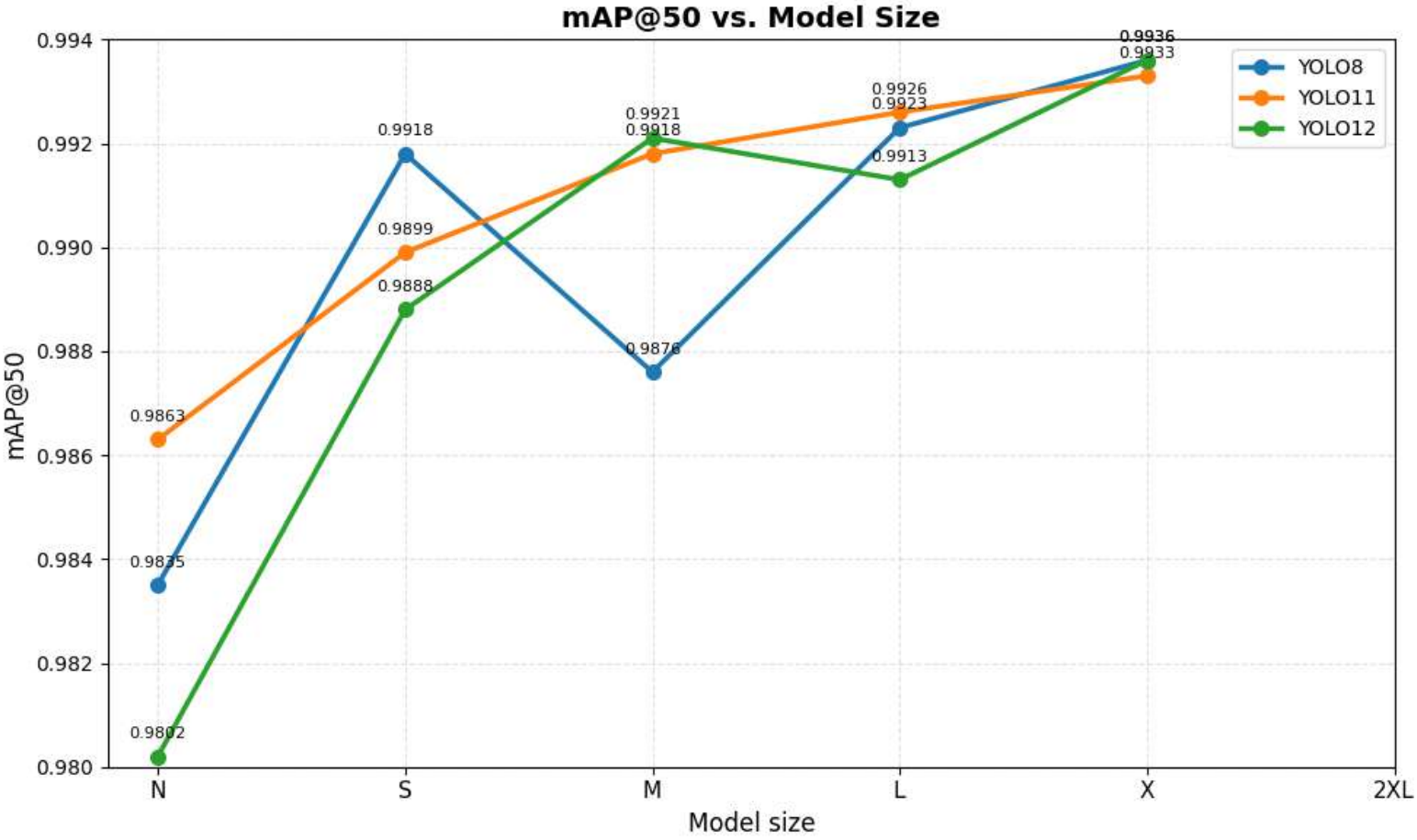
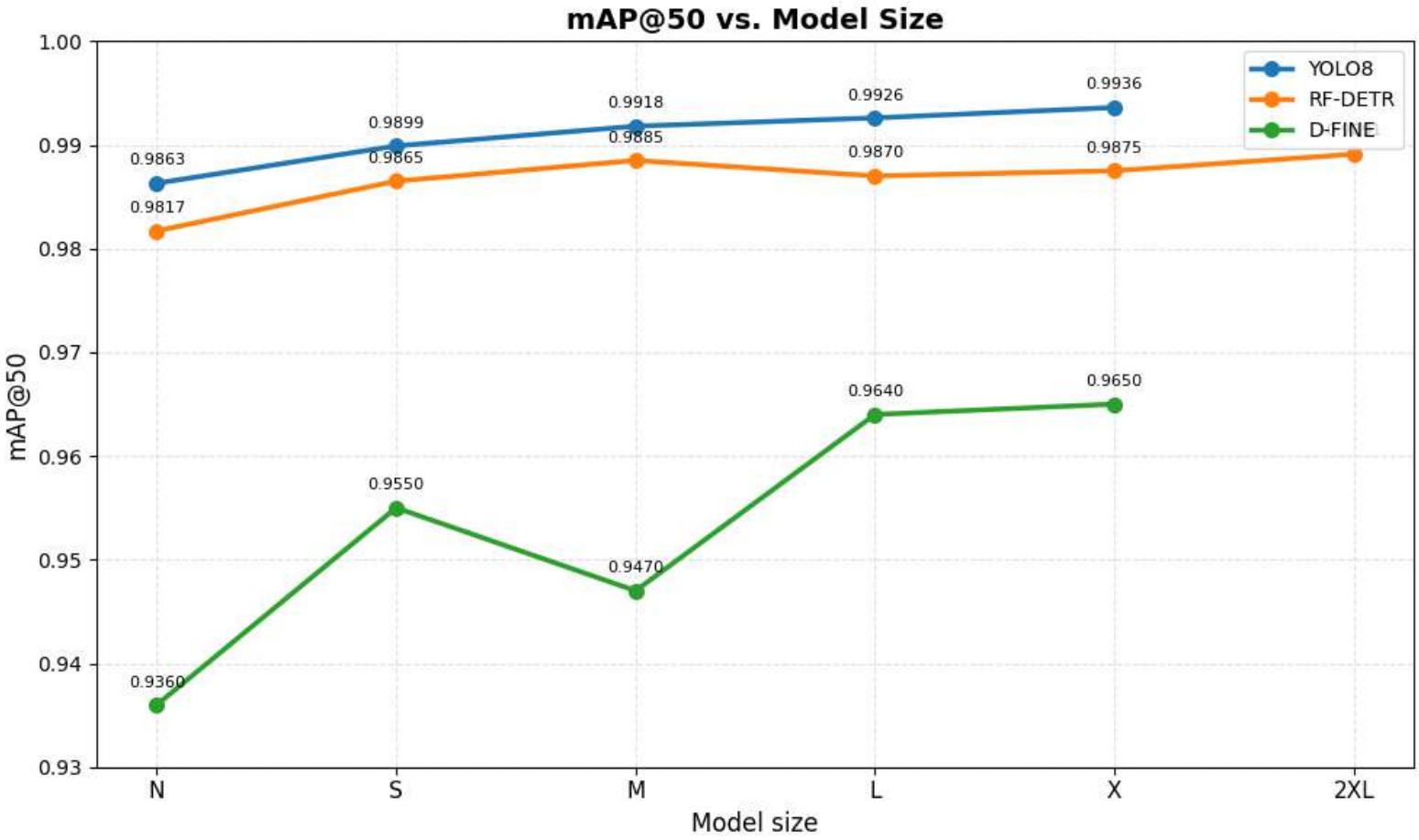
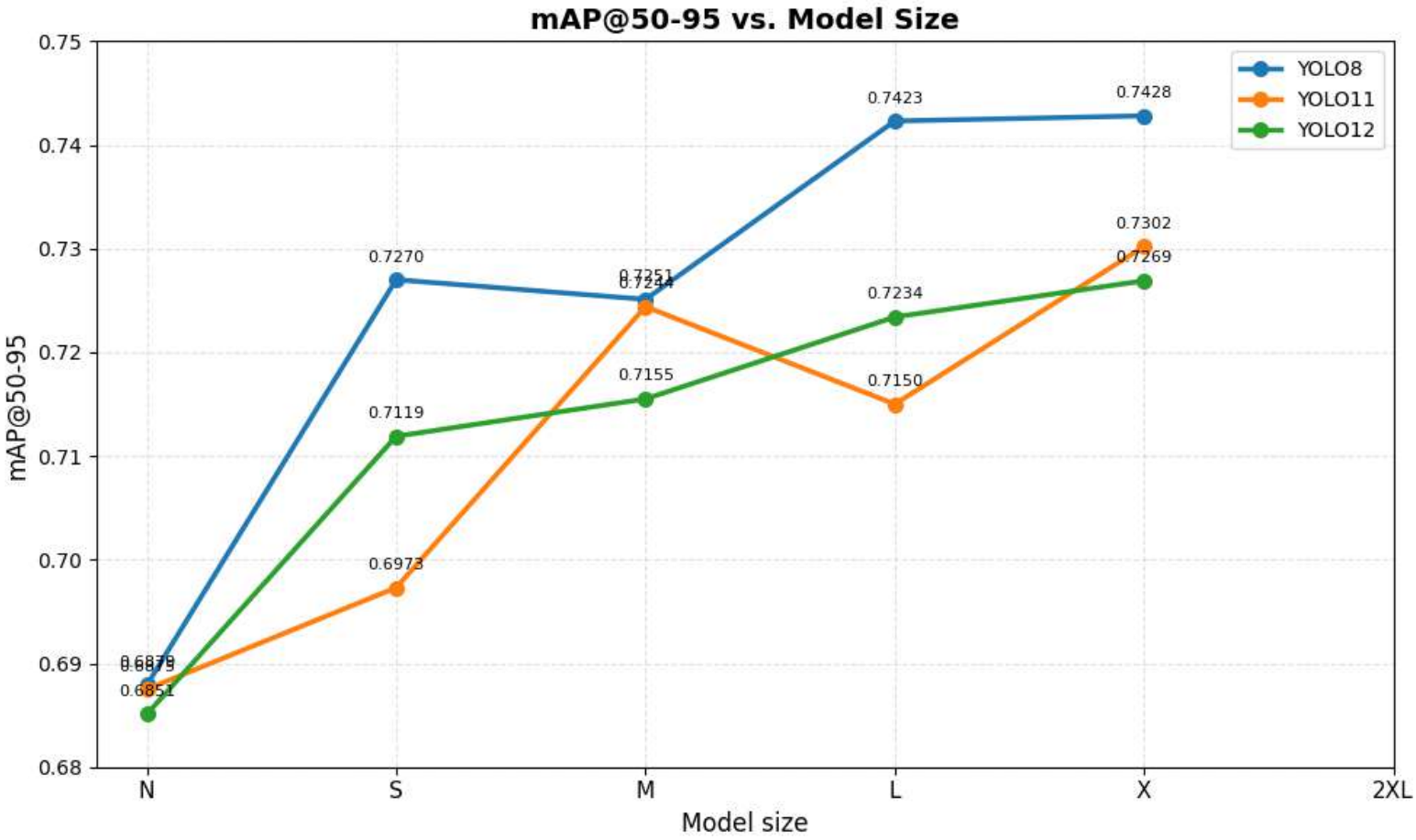
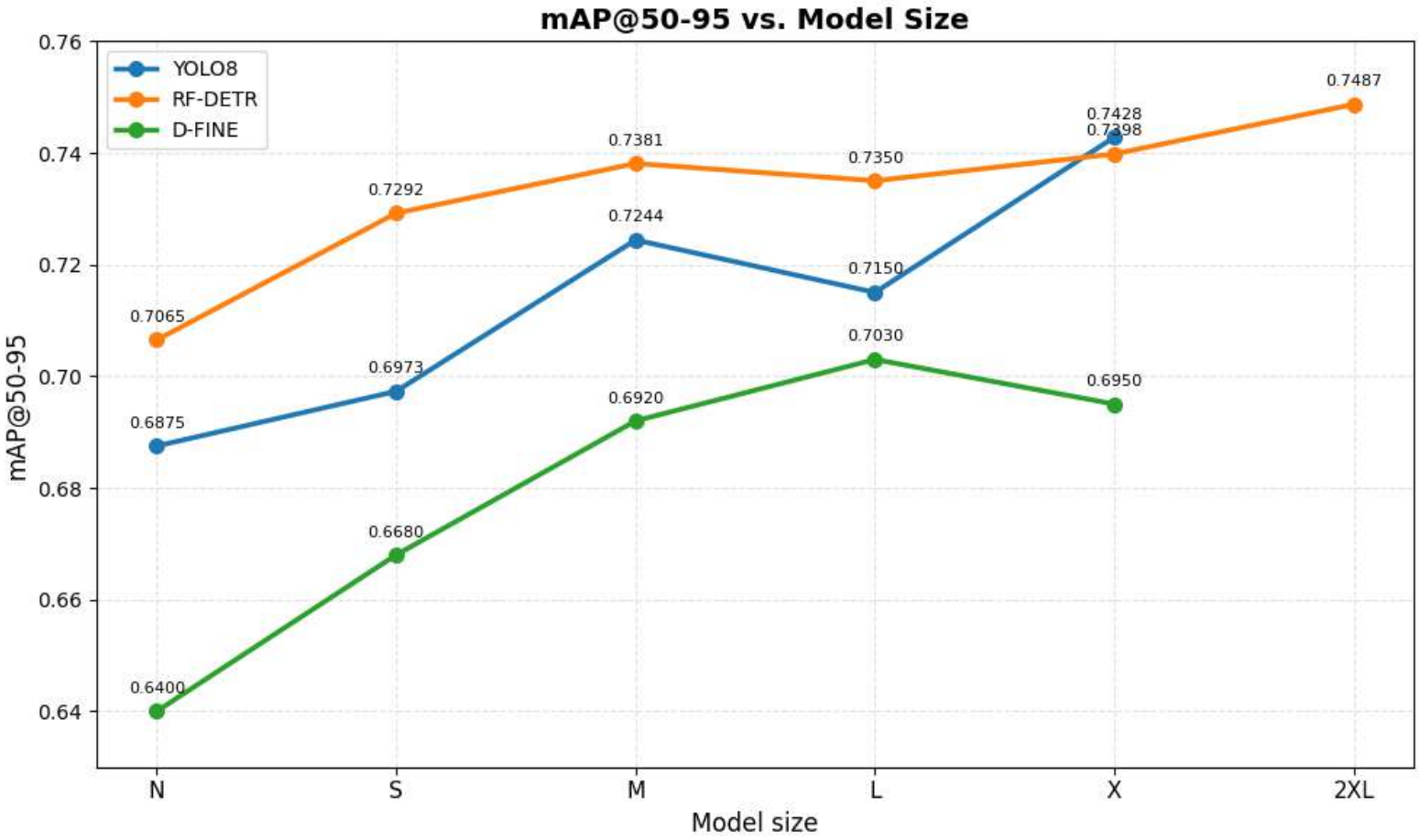
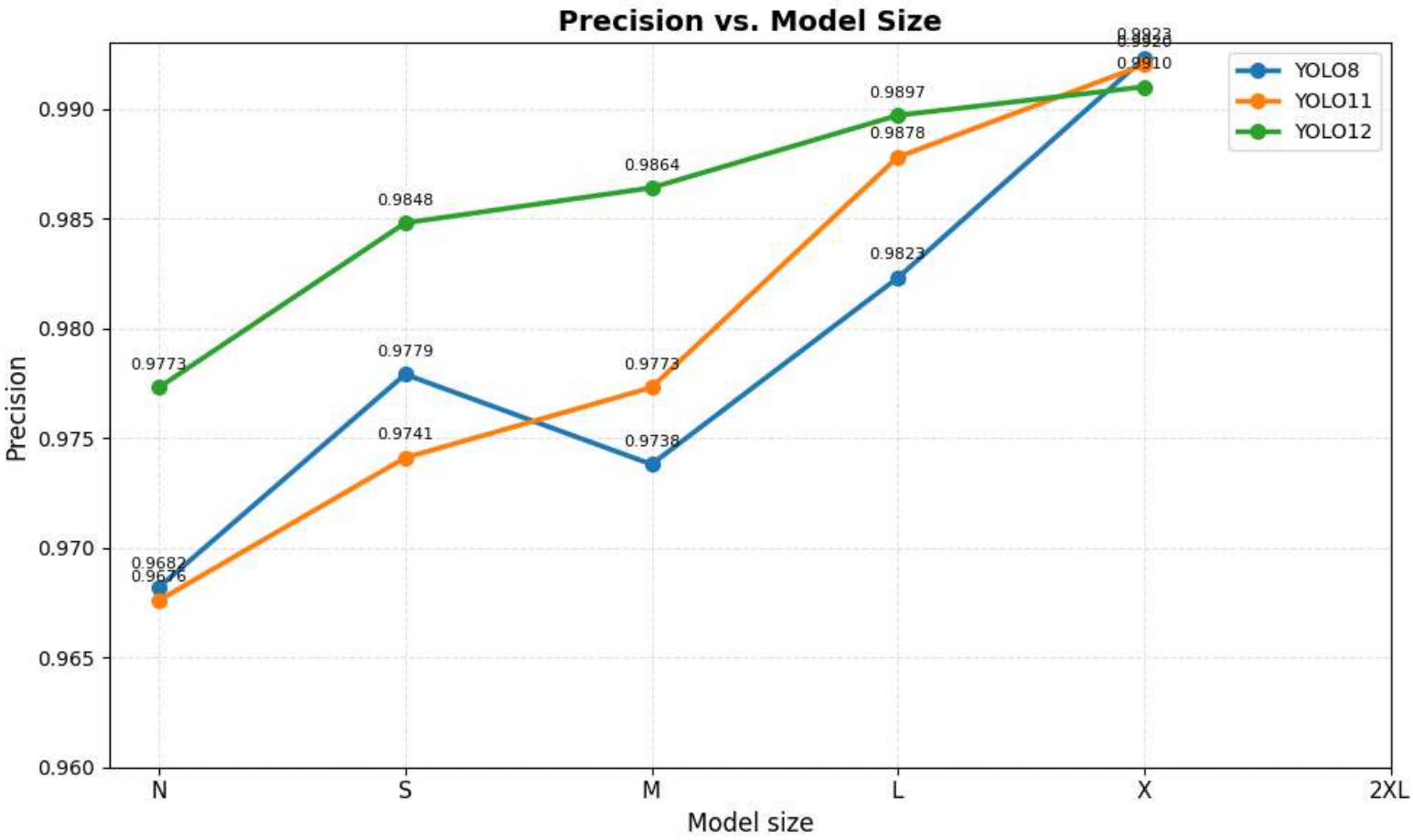

Number of Parameters vs. Model Size
YOLO8
YOLO11
YOLO12
RF-DETR
D-FINE
Parameters (M)
Model size
N
S
M
L
X
2XL
20
40
60
80
100
120
30.50
32.10
33.70
33.90
31.00
126.40
126.90
43.69
68.23
62.00
59.21
56.97
25.90

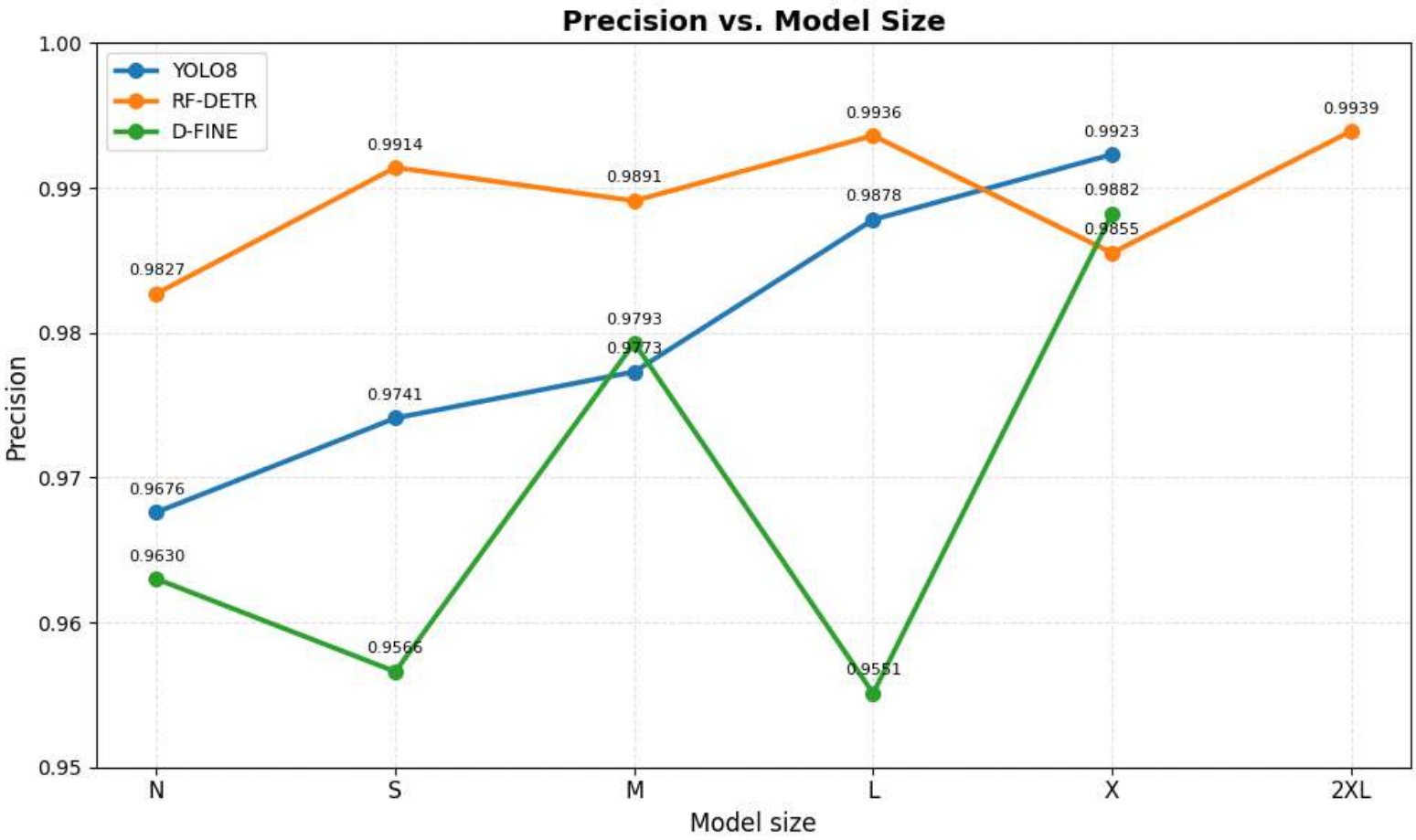

mAP@50 vs. Model Size
mAP@50
Model size
N
S
M
L
X
2XL
0.994
0.992
0.990
0.988
0.986
0.984
0.982
0.980
YOLO8
YOLO11
YOLO12
0.9863
0.9899
0.9888
0.9835
0.9918
0.9876
0.9921
0.9918
0.9802
0.9926
0.9923
0.9913
0.9936
0.9933

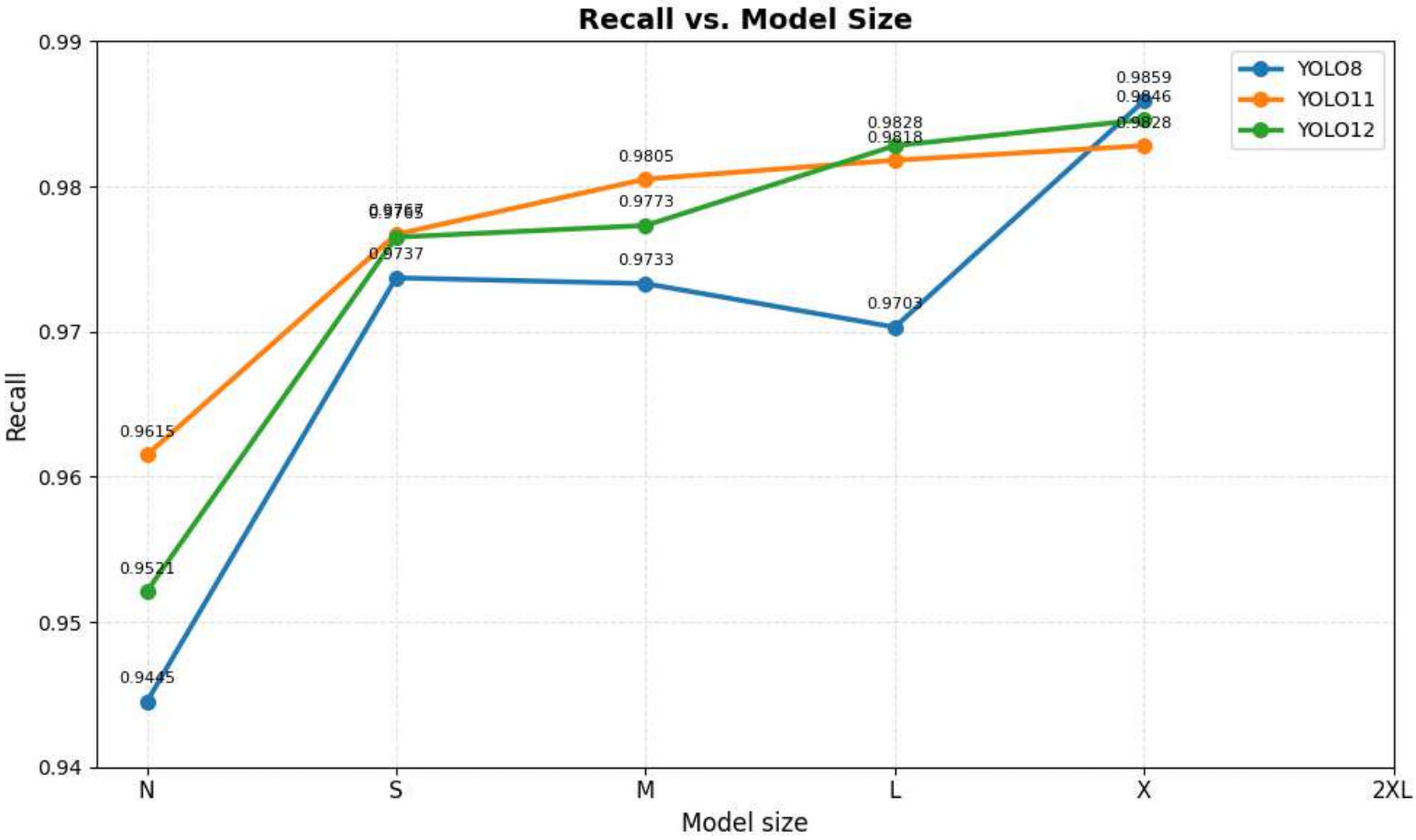

mAP@50 vs. Model Size
YOLO8
RF-DETR
D-FINE
mAP@50
Model size
N
S
M
L
X
2XL
1.00
0.99
0.98
0.97
0.96
0.95
0.94
0.93
0.9863
0.9899
0.9918
0.9926
0.9936
0.9817
0.9865
0.9885
0.9870
0.9875
0.9360
0.9550
0.9470
0.9640
0.9650

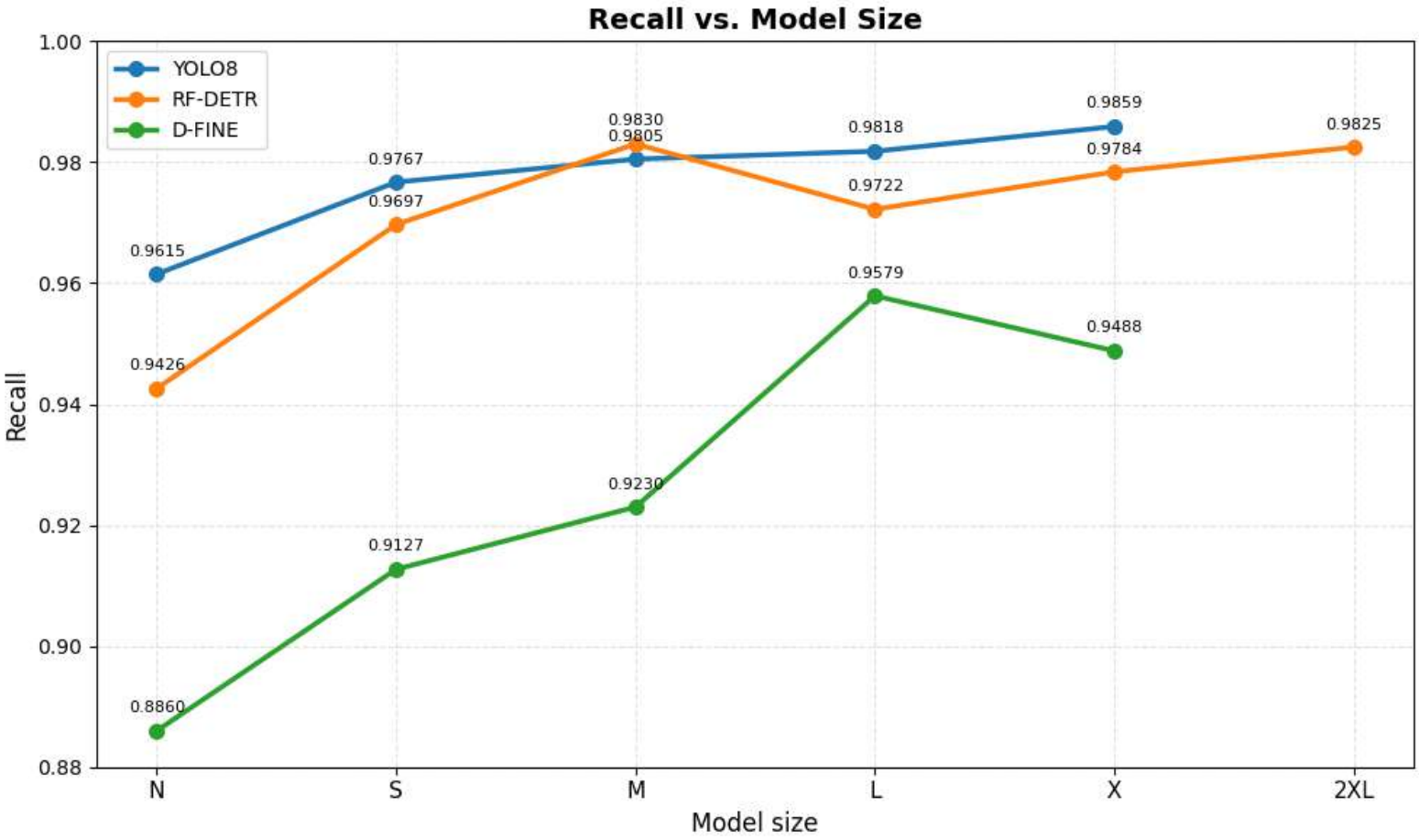

mAP@50-95 vs. Model Size
YOLO8
YOLO11
YOLO12
mAP@50-95
Model size
N
S
M
L
X
2XL
0.75
0.74
0.73
0.72
0.71
0.70
0.69
0.68
0.7423
0.7428
0.7270
0.7251
0.7244
0.7302
0.7269
0.7234
0.7155
0.7150
0.7119
0.6973
0.6879
0.6851

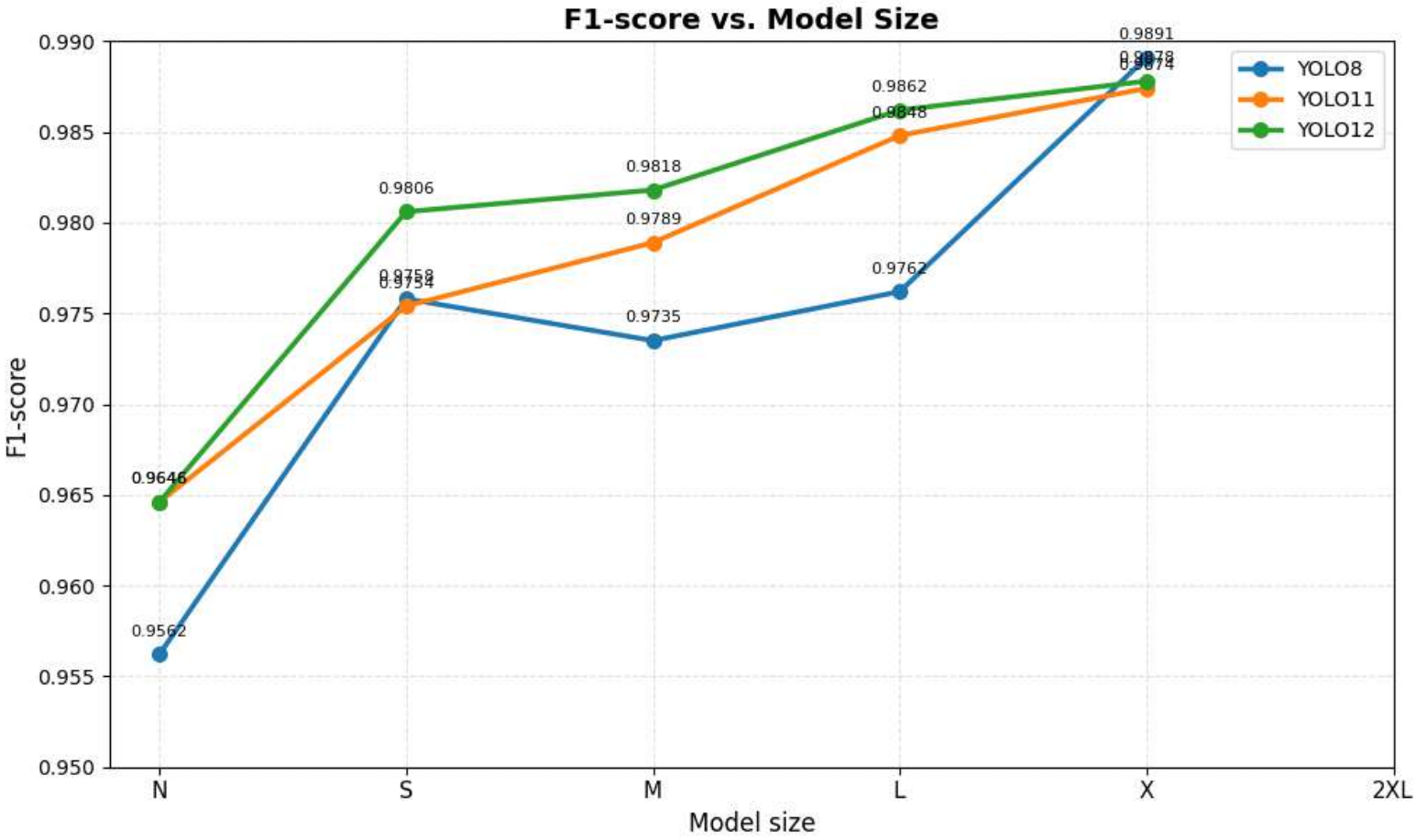

mAP@50-95 vs. Model Size
YOLO8
RF-DETR
D-FINE
mAP@50-95
Model size
N
S
M
L
X
2XL
0.76
0.74
0.72
0.70
0.68
0.66
0.64
0.7065
0.7292
0.7381
0.7350
0.7398
0.7487
0.6875
0.6973
0.7244
0.7150
0.7428
0.6400
0.6680
0.6920
0.7030
0.6950

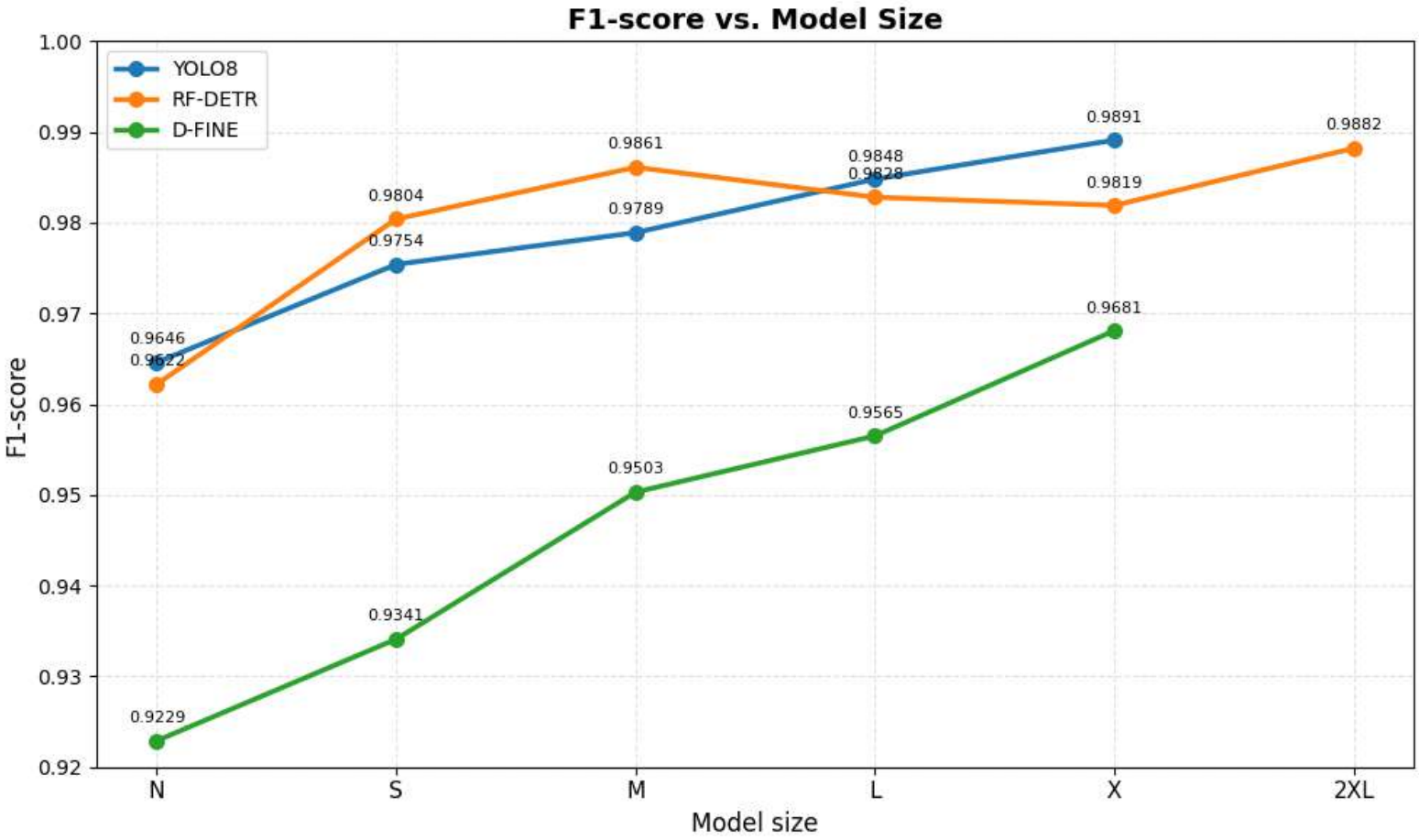

Precision vs. Model Size
YOLO8
YOLO11
YOLO12
Precision
Model size
N
S
M
L
X
2XL
0.960
0.965
0.970
0.975
0.980
0.985
0.990
0.9682
0.9676
0.9779
0.9741
0.9738
0.9823
0.9878
0.9910
0.9773
0.9773
0.9848
0.9864
0.9897


Precision vs. Model Size
YOLO8
RF-DETR
D-FINE
Precision
Model size
1.00
0.99
0.98
0.97
0.96
0.95
N
S
M
L
X
2XL
0.9827
0.9914
0.9891
0.9936
0.9855
0.9939
0.9676
0.9741
0.9773
0.9878
0.9923
0.9630
0.9566
0.9793
0.9551
0.9882


Recall vs. Model Size
YOLO8
YOLO11
YOLO12
Recall
Model size
0.99
0.98
0.97
0.96
0.95
0.94
N
S
M
L
X
2XL
0.9615
0.9571
0.9445
0.9737
0.9733
0.9703
0.9805
0.9773
0.9828
0.9818
0.9859
0.9846


Recall vs. Model Size
YOLO8
RF-DETR
D-FINE
Recall
Model size
1.00
0.98
0.96
0.94
0.92
0.90
0.88
N
S
M
L
X
2XL
0.9615
0.9767
0.9830
0.9818
0.9859
0.9426
0.9697
0.9805
0.9722
0.9784
0.9825
0.8860
0.9127
0.9230
0.9579
0.9488


F1-score vs. Model Size
F1-score
Model size
0.990
0.985
0.980
0.975
0.970
0.965
0.960
0.955
0.950
N
S
M
L
X
2XL
YOLO8
YOLO11
YOLO12
0.9891
0.9862
0.9848
0.9818
0.9806
0.9789
0.9762
0.9758
0.9754
0.9735
0.9646
0.9562


F1-score vs. Model Size
YOLO8
RF-DETR
D-FINE
F1-score
Model size
1.00
0.99
0.98
0.97
0.96
0.95
0.94
0.93
0.92
N
S
M
L
X
2XL
0.9646
0.9622
0.9754
0.9789
0.9848
0.9828
0.9891
0.9804
0.9861
0.9819
0.9882
0.9229
0.9341
0.9503
0.9565
0.9681